

sfully displayed the programmed pattern . 2. different patterns can be created by changing the binary values in the code . 3. the matrix can display symbols ,letters, numbers, and simple animations .

LEARNING OUTCOME

1. learned how to interface an LED matrix with arduino
2. understood row and columns addressing in LED matrices.
3. Gained experience in creating visual patterns using programming

PROBLEAMS

pattern was not displayed correctly .

SOLUTION:

Checked wiring connections and verified the binary pattern data in the coe .

FUTURE SCOPE

1. Display scrolling text .
2. create animations and games
3. Design digital notice boards

REFERENCES

1. Arduino official documentation
2. max 7219 led matrix datasheet
3. Arduino LED matrix library documentation

TABLE OF CONTENTS

1.TITLE 2 . AIM 3. OBJECTIVE 4. COMPONENTS REQUIRED 5. CIRCUIT DIAGRAM 6. PRODUCER 7. ADVANTAGE 8. CONCLUSION

PROJECT REPORT ON 555 TIME IC

TITLE

design and implementation of a 555 timer circuit

AIM

TO study the working of the 555 timer IC and generate timing pulses using an astable multivibrator circuit

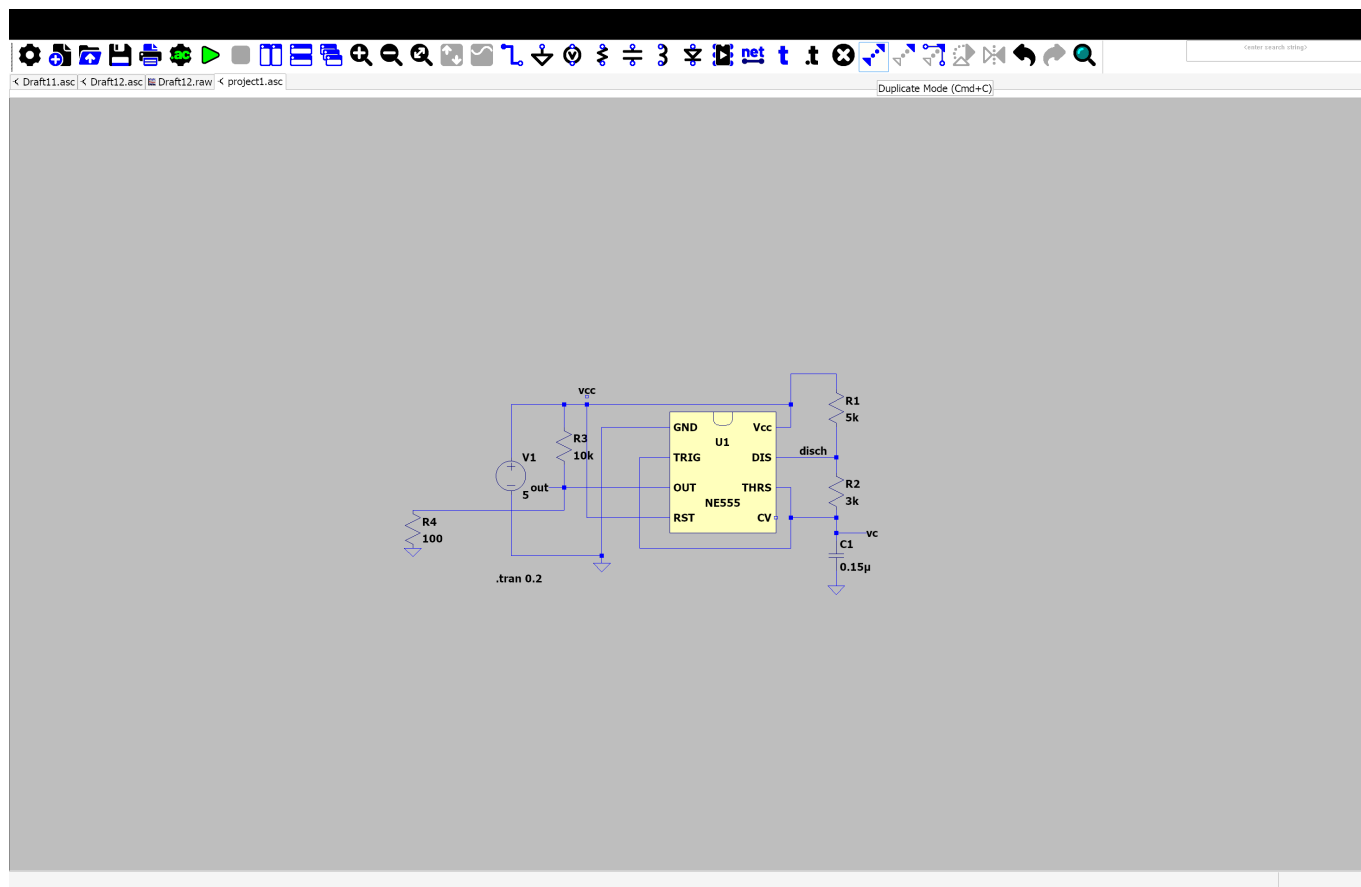
OBJECTIVE

1. TO understand the operation of the 555 time IC
2. To generate a continuous square wave output
3. To observe the charging and discharging of a capacitor

COMPONENTS REQUIRED

5. 555timer IC
6. Resistor r1 (1kohm)
7. Resistor R2(10k ohm) capacitor c1 (10uf) connecting wires

CIRCUIT DIAGRAM



PROCEDURE

1. Place the timer IC on the breadboard
2. connect the power supply.
3. turn on the power supply
4. observe the blinking led.

Observation

6. The LED blinking rate depends on the resistor and capacitor value
7. The LED blinks continuously
8. A square wave is obtained at the output pin .

APPLICATION

9. LED flashers
10. pulse generators clock circuit alarm system

ADVANTAGES

11. low
12. easy to use
13. reliable operation

CONCLUSION

The project demonstrated the working principle of the 555 timer IC the circuit successfully generated periodic pulses and showed how resistor and capacitor values affect the timing characteristic of the output signal .

INVERTING AND NON INVERTING AMPLIFIER USING OP AMP

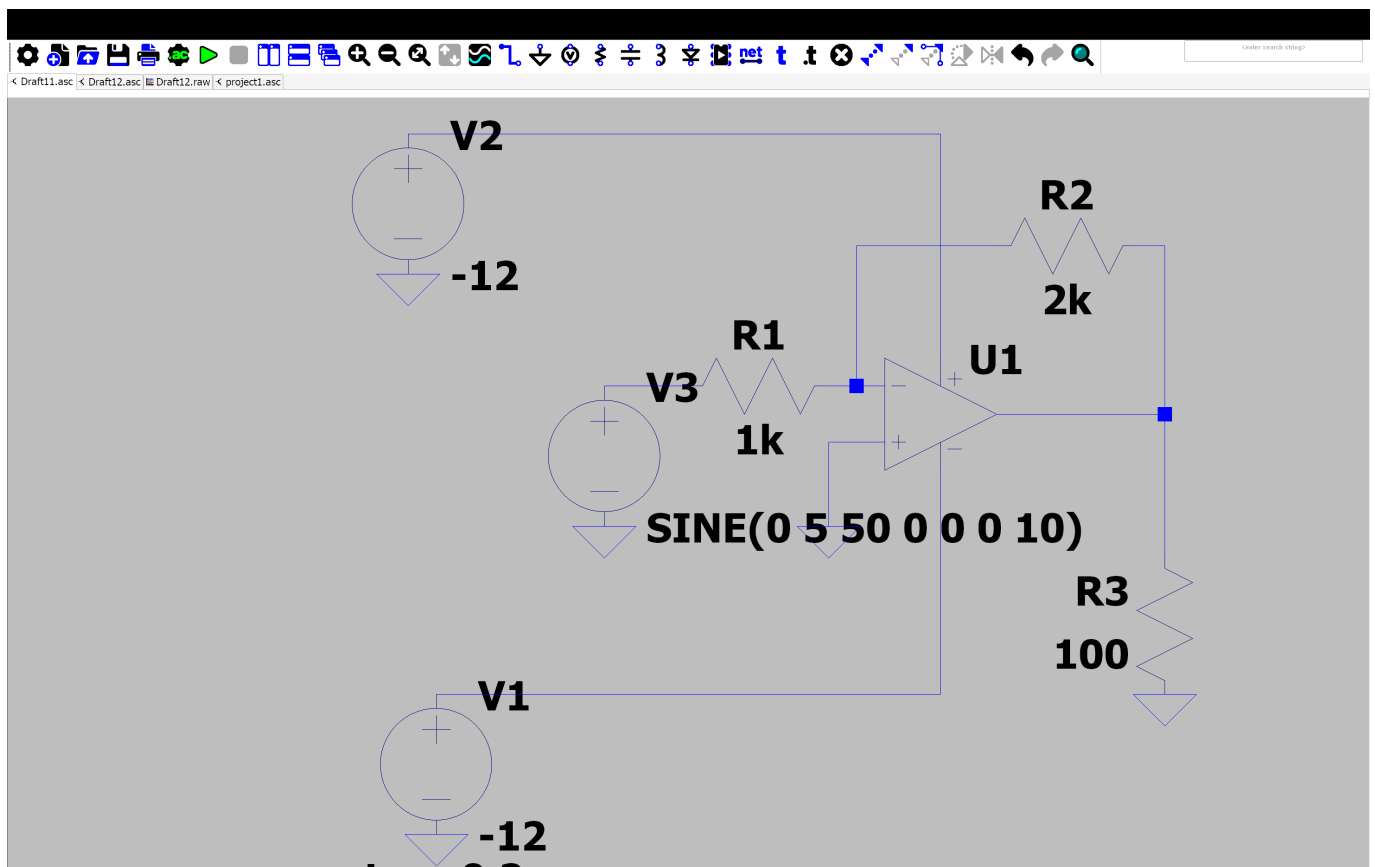
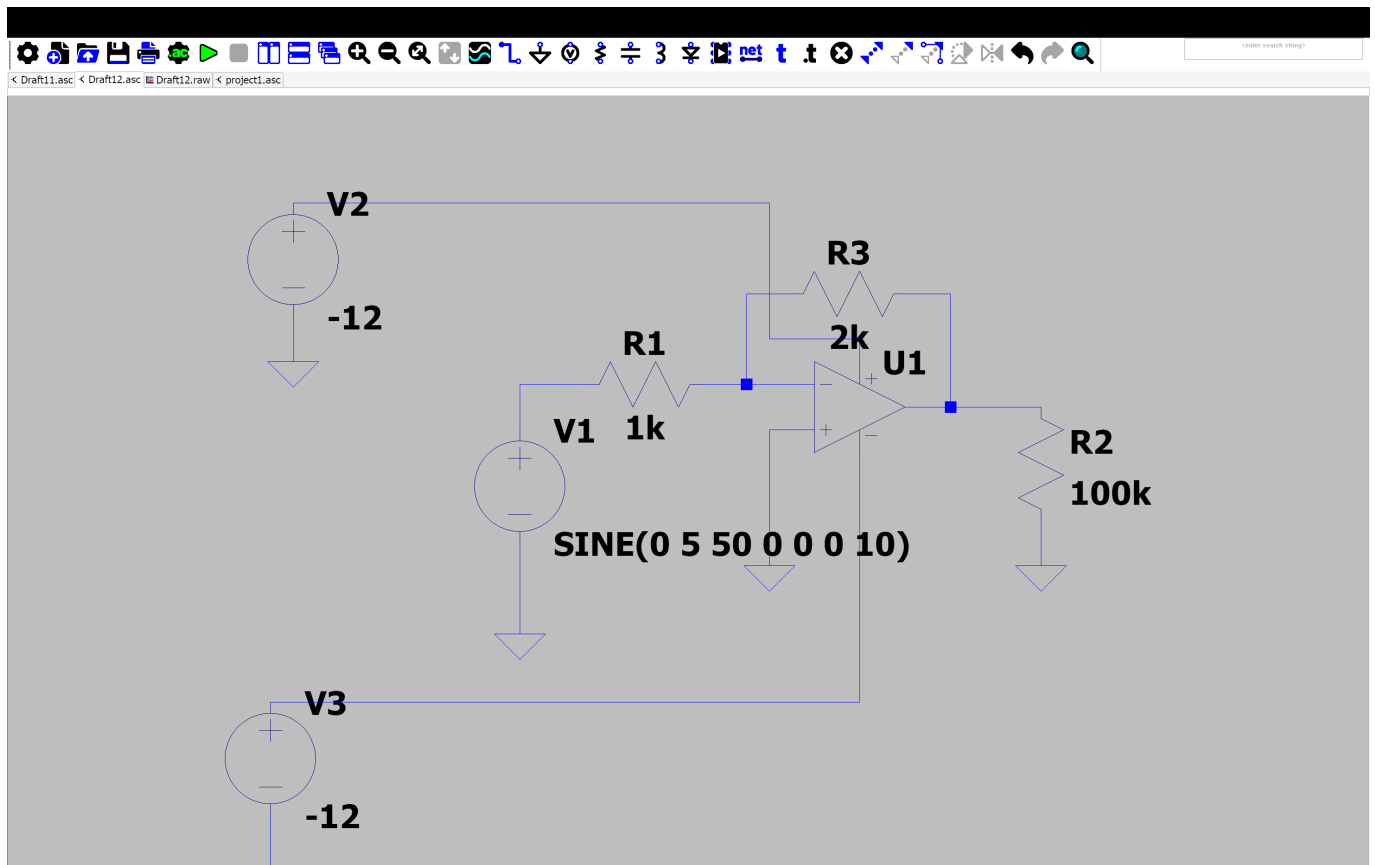
AIM

To study and analyze the operation of inverting and non inverting amplifier using an operational amplifier (op amp)

Objective

1. to understand the working principle of inverting and non inverting amplifiers
2. To calculate and verify the voltage gain of both amplifier configurations
3. To compare their characteristics and applications .

CIRCUIT DIAGRAM INVERTING AND NON-INVERTING



Right click to edit "-12", the Value of V2

APPLICATION

INVERTING AMPLIFIER

1. Signal conditioning

2. Audio amplifiers

3. active filters

NON INVERTING AMPLIFIER

1. Voltage follows

2. sensor signal amplification

3. instrumentation circuits.

TYPING REPORT

sno	DATE	TYPING SCORE
1.	2/6/26	24
2.	3/6/26	24
3.	4/6/26	24
4.	5/6/26	22
5.	6/6/26	23
6.	7/6/26	26
7.	8/6/26	30
8.	9/6/26	30
9.	10/6/26	28
10	11/6/26	23
11.	12/6/26	26
12.	13/6/26	27
13.	14/6/26	27
14	15/6/26	26
15	16/6/26	26
16	17/6/26	25
17	18/6/26	26
18.	19/6/26	25
19.	20/6/26	26
20.	21/6/26	27
21.	22/6/26	27
22	23/6/26	26
23	24/6/26	26

sno	DATE	TYPING SCORE
24	25/6/26	25
25	26/6/26	26
26.	27/6/26	25
27.	28/6/26	27
28.	29/6/26	27
29	30/6/26	30
30	1 /7/26	30
31	2/7/26	29
32	3/7/26	30
33.	4/7/26	27
34	5/6/26	29
35	6/6/26	29
36	7/6/26	30

HTML CODE WITH SQL DATABASE

AIM

To create a simple web page using HTML and store user information in an SQL database .

OBJECTIVE

1. To design a user input form using HTML.
2. To store the entered data in an SQL database .
3. To retrieve and display records from the database .

SOFTWARE REQUIRED

1. Notepad
2. XAMPP SERVER
3. MYsql Database
4. SQL

INTRODUCTION

HTML (hyper text markup language) is used to create web pages and collect user input through forms .SQL (Structure Query language) is used to create and manage database .since HTML cannot communicate directly with a database , a server- side language such as PHP is used as an intermediary between HTML and SQL .this project demonstrates a simple student registration system where student can enter their details ,and the information is stored in a MYSQL database .

Software and Hardware Requirement

1. Operating system : windows 10/11
2. Text editor :NOTEPAD
3. XAMPP server
4. MySQL database

HARDWARE REQUIREMENTS

1. Computer or laptop
2. minimum 4GB RAM
3. 500 MB free dissk space

THEORY

HTML

HTML is the standard markup language used for creating web pages . it provides various elements such as forms , text boxes, button , and labels for user interaction .

SQL :

SQL is a language used to manage relational databasae .It allows user to create databases, table , insert record ,update data , and retrieve information .

PHP :

PHP is a server -side scripting language thatbprocesses from data and interacts with the MYSQL database

FLOW OF THE SYSTEM

1. User opens the registration page .
2. user enters details in the html form.
3. the form sends data to the php file .
4. php establishes a connection with MYSQL .
5. SQL query inserts the data into the database .
6. Asuccess message is displayed .

HTML CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Page</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      background-color: #f4f4f4;
      margin-top: 50px;
    }

    h1 {
      color: #333;
    }

    button {
      padding: 10px 20px;
      font-size: 16px;
      background-color: #4CAF50;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }

    button:hover {
      background-color: #45a049;
    }
  </style>
```

```
</head>
<body>

  <h1>Welcome to My Website</h1>
  <p>This is a simple HTML page.</p>

  <button onclick="showMessage()">Click Me</button>

  <script>
    function showMessage() {
      alert("Hello! Welcome to my website.");
    }
  </script>

</body>
</html>
```

ADVANTAGE

1. EASY Data management
2. Reduces paperwork
3. quick retrieval of records
4. Improves accuracy and efficiency .

APPLICATION

1. School management systems
2. college recorg systems
3. Employee management systems
4. Library management systems

OUTPUT

The user enters the details in the HTML form,and the information is stored in the SQL datadase table .

CONCLUSION

The experiment sucessfully demonstrated how HTML can be used to collect user data and SQL can be used to store and manage that data in a database .

BMS BUTTON

AIM

To study the operation of a push button and control an output device using a BMS (Basic Monitoring/Management System).

OBJECTIVE

To understand the working of push buttons and their application in automation and control systems.

THORY

A Button Control is a graphical user interface (GUI) element used to execute a command when clicked by the user. It is one of the most commonly used controls in BMS applications. Button controls help users interact with software easily and efficiently.

SOFTWARE REQUIRED

1. BMS Software
2. COMPUTER SYSTEM

PROCEDURE

1.open the BMS software 2. create a new project. 3. Drag and drop a Button Control onto the form. 4. Change the button text to "Click Me". 5. Write code for the button click event. 6. Run the application. 7. Click the button and observe the result.

PROGRAM CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>BMS Dashboard Sketch</title>
  <style>
    body {
      font-family: 'Courier New', Courier, monospace;
      background-color: #ffffff;
      padding: 30px;
    }
    .bms-card {
      border: 2px solid #000000;
      padding: 15px;
      width: 320px;
      background: #ffffff;
      margin-bottom: 5px;
    }
  </style>
</head>
<body>
  <div class="bms-card">
    <button type="button">Click Me</button>
  </div>
</body>
</html>
```

```
}  
.bms-card h3 {  
    margin: 0 0 10px 0;  
    font-size: 18px;  
    letter-spacing: 1px;  
}  
.info-text {  
    margin: 5px 0;  
    font-size: 15px;  
}  
.top-btns {  
    margin: 10px 0;  
}  
.small-btn {  
    border: 1px solid #000;  
    background: none;  
    padding: 2px 8px;  
    margin-right: 5px;  
    font-size: 12px;  
}  
.btn-section {  
    margin: 15px 0;  
}  
.row {  
    margin-bottom: 8px;  
    display: flex;  
    align-items: center;  
}  
.clickable-btn {  
    border: 2px solid #000000;  
    background: #ffffff;  
    padding: 5px 15px;  
    margin-right: 8px;  
    cursor: pointer;  
    font-weight: bold;  
    font-family: inherit;  
}  
.clickable-btn:hover {  
    background: #f0f0f0;  
}  
.arrow-label {  
    font-size: 14px;  
}  
.arrow-down {  
    font-size: 24px;  
    margin-left: 20px;  
    margin-top: -5px;  
    margin-bottom: 5px;  
}  
.details-box {  
    border: 2px solid #000000;  
    padding: 12px;  
    width: 200px;  
    background: #ffffff;
```

```

        margin-left: 10px;
    }
    .details-box p {
        margin: 4px 0;
        font-size: 14px;
    }
</style>
</head>
<body>

<div class="bms-card">
    <h3>BMS UI</h3>
    <div class="info-text">Total Volt: 0.8V</div>
    <div class="info-text">Connection: 2S (3.2V - 4.4V)</div>

    <div class="top-btns">
        <button class="small-btn">C1</button>
        <button class="small-btn">C2</button>
    </div>x

    <div class="info-text">No. of Cells: 2</div>

    <div class="btn-section">
        <div class="row">
            <button class="clickable-btn">C1</button>
            <button class="clickable-btn">C2</button>
            <span class="arrow-label">&arr; clickable button</span>
        </div>
        <div class="row">
            <button class="clickable-btn">C3</button>
            <button class="clickable-btn">C4</button>
            <button class="clickable-btn">C5</button>
        </div>
    <button class="clickable-btn">C6</button>
    </div>
</div>

<div class="arrow-down">&ShortDownArrow;</div>

<div class="details-box">
    <p><b id="display-name">C1</b></p>
    <p>Battery Volt: <span id="display-volt">---</span></p>
    <p>Temp: <span id="display-temp">---</span></p>
    <p>Current State: <span id="display-state">---</span></p>
</div>

<script>
    const dataMap = {
        'C1': { volt: '3.6V', temp: '26°C', state: 'Normal' },
        'C2': { volt: '3.4V', temp: '28°C', state: 'Normal' },
        'C3': { volt: '1.1V', temp: '45°C', state: 'Hot' },
        'C4': { volt: '0.0V', temp: '24°C', state: 'Offline' },
        'C5': { volt: '3.5V', temp: '27°C', state: 'Normal' },
        'C6': { volt: '3.3V', temp: '26°C', state: 'Normal' }
    }

```

```
};

document.querySelectorAll('.clickable-btn').forEach(btn => {
  btn.addEventListener('click', function() {
    const name = this.innerText.trim();
    const info = dataMap[name];

    if (info) {
      document.getElementById('display-name').innerText = name;
      document.getElementById('display-volt').innerText = info.volt;
      document.getElementById('display-temp').innerText = info.temp;
      document.getElementById('display-state').innerText =
info.state;
    }
  });
});
</script>
```

SCREENSHORT

- → ↻ ⓘ 127.0.0.1/bms_project/index.php

BMS UI

Total Volt: 0.8V

Connection: 2S (3.2V - 4.4V)

No. of Cells: 2

→ clickable button



C3

Battery Volt: 1.1V

Temp: 45°C

Current State: Hot

OBSERVATION

1. The button appeared on the form. Clicking the button 2. displayed a message box.
2. The button executed the assigned command correctly. Result

RESULT

The Button Control was created successfully and performed the desired action when clicked.

CONCLUSION

The Button Control is an important GUI component used to interact with applications. The experiment demonstrated how a button can be created and programmed to perform specific tasks